

roomscan — technical report

Phone capture → dimensioned, stitched floor plan with damage scope. All numbers are regenerated by python bench/run_benchmark.py and written into this report by python bench/make_report.py (run of 2026-10-05 05:21:27).

0. Scope and the data constraint

We were given three Stray Scanner captures of one flat (LiDAR depth, ARKit poses, intrinsics, RGB video) and no iPhone, no access to the property and no laser. Consequences, stated once and honoured throughout: - **No ground truth**. LiDAR numbers are reported as *repeatability* between three captures of the same rooms (proven to be one flat by wall-map correlation, DECISIONS D6); video/photo numbers are reported against the LiDAR tier of the same capture. Nothing is called “accuracy” that is really agreement. - **Thin tiers are derived**. The video tier gets only rgb.mp4 (upright, metadata stripped); the photo tier gets 2–8 EXIF-free stills per room picked the way the protocol tells a person to shoot. ARKit poses are never given to them, but are used *afterwards* to score the video tier’s camera path (§5). - **Damage is staged digitally** onto real wall planes and re-rendered through the capture’s own poses (§6). - **Head-to-head** with a consumer app needs that app run in the same rooms: not possible here (§8).

1. Architecture

capture → tier front-end → Scene → layout → rooms/walls/openings → damage → JSON + plan
lidar | video | photo (points, oriented normals, camera views, error model)

Every tier only has to produce a Scene; everything downstream is shared code, which is what makes “same output contract from every tier” true by construction (scene.py, analyze.py, export.py). - **Frame**: gravity from ARKit (LiDAR) or from the floor normal (video/photo); Manhattan yaw from the circular histogram of wall normals; plan coordinates (u, v, h above floor). - **Free space**: per view, a 2-D visibility fan (camera → farthest return per 1° azimuth); union over views. - **Walls**: cells whose vertical points span > 0.9 m of height; Manhattan wall-face segments by 1-D peak finding. - **Rooms**: interior = free – walls – doorway closures; watershed on the distance transform seeded by cores that survive a 0.42 m erosion; room geometry re-snapped to the wall-line arrangement (D4, D13, fix loop). - **Wall faces**: for each polygon edge, the outermost strong, floor-reaching, *room-facing* surface layer within 0.5 m, fitted from raw points (robust median), never from the 2 cm raster. - **Openings**: 2 cm bins along each face; a run with no surface points in the door band *and* see-through evidence beyond the wall; widths from jamb points (sub-bin). A gap without see-through is “wall not observed”, never an opening — that is the phantom-opening guard. - **Ceiling**: highest strong horizontal layer inside the room polygon, plane-refined; if the capture never looked up, the value is a *prior* anchored at the highest observed wall point and marked observed: false.

2. Tier design and device matrix

tier	front-end	scale source	intervals (1σ model, tiers/*.py)
LiDAR	Stray loader; RGB/depth frame offset estimated (D9); keyframes 10 cm / 8°	LiDAR	face 4 mm, scale 0.3%, drift 0.4 mm per m walked
video	KLT chains at 10 fps between depth keyframes, PnP-RANSAC with metric mono-depth; fast turns re-decoded at native rate; ICP / constant-velocity fallback; focal from Manhattan vanishing points (709 vs 711 px true)	mono-depth (Depth Anything V2 metric-indoor) × device scale calibrated on LiDAR (0.766)	face 30 mm, scale 5%, drift 1.5%/m walked (measured, D24)
photo	per photo: metric depth → gravity + Manhattan frame; 4 yaws × 2-D shift solved by FFT top-view correlation; rooms joined through doorway shots	same as video	face 40 mm, scale 5%

Hardware: see docs/DEVICE_MATRIX.md (LiDAR tier needs a Pro iPhone; video/photo any iPhone 15+; processing on a 2-core CPU laptop).

3. Drift

ARKit is locally excellent but drifts over a multi-room walk. We cut the keyframes into ~1.5 m fragments, add ARKit odometry edges between consecutive fragments and **loop-closure edges** between non-consecutive overlapping fragments from point-to-plane ICP, projected to 4-DoF (yaw + translation; gravity is trusted), and optimise the pose graph (Open3D LM with line-process outlier pruning). Same module serves the video tier.

capture	drift correction	footprint m²	wall sharpness	loop residual (cm) before → after	loop edges kept
single_room	on	23.12	0.6998	8.4 → 2.7	4
single_room	off (ARKit as-is)	23.86	0.6065	—	—
floor_only	on	63.80	0.489	14.5 → 6.1	14
floor_only	off (ARKit as-is)	64.25	0.5145	—	—
with_ceiling	on	68.10	0.4625	10.2 → 3.3	68
with_ceiling	off (ARKit as-is)	70.77	0.3541	—	—

Cross-capture effect (floor_only vs with_ceiling): registration score 0.331 with correction vs 0.297 without; walls within 3%: 29.2% vs 16.7%; median |ΔL| 31.15 vs 31.75 cm.



with_ceiling stitched plan, drift correction off (left) and on (right)

4. Error budget

Every Measure carries error_budget_1sigma: wall length = face fit (each end) ⊕ sensor face bias ⊕ scale × L ⊕ drift × path; area = $\sum (L_i \cdot \sigma_{\text{face},i})$ ⊕ scale ⊕ drift; ceiling = floor-plane fit ⊕ ceiling-plane fit ⊕ sensor ⊕ scale; opening width = jamb spread ⊕ sensor ⊕ scale. Unobserved quantities get priors with observed: false.

5. Results



Stitched LiDAR plans: floor_only, with_ceiling, single_room (one command each)

LiDAR repeatability (same tier, different captures of the same rooms)

pair	room pairs	walls	median $ \Delta L $ cm	within 1 cm/0.5%	within 3%	within 8%	ref in 95% CI	openings matched/missed/phantom	opening ≤ 2 cm
floor_only~with_ceiling	5	24	31.15	4.2%	29.2%	41.7%	45.8%	3/5/7	6.7%
with_ceiling~single_room	2	7	25.3	28.6%	42.9%	42.9%	42.9%	1/4/2	14.3%
floor_only~single_room	2	6	11.95	16.7%	50.0%	66.7%	66.7%	0/1/3	0.0%

Thin tiers vs the LiDAR tier of the same capture

comparison	rooms LiDAR/tier	footprint err % (gate: video ± 3 , photo ± 8)	LiDAR footprint in 95% CI	room overlap m²	walls	median $ \Delta L $ %	walls within 3%	walls within 8%	camera path ATE m
video_vs_lidar:single_room	4/1	38.5	True	0.0	0	—	—	—	1.099
photo_vs_lidar:single_room	4/3	-69.3	False	0.3735	0	—	—	—	—
video_vs_lidar:floor_only	5/3	-25.8	True	0.1505	5	60.87	0.0%	20.0%	1.466
photo_vs_lidar:floor_only	5/3	-69.0	False	0.2733	0	—	—	—	—
video_vs_lidar:with_ceiling	5/2	-6.5	True	0.5219	0	—	—	—	4.816
photo_vs_lidar:with_ceiling	5/3	-81.4	False	0.0	0	—	—	—	—

Walls are compared only inside room pairs overlapping with IoU ≥ 0.4 (— = no such pair). Camera path ATE: RMS distance between the video tier's RGB-only path and ARKit after rigid alignment.

Ceiling heights (only with_ceiling observes ceilings)

room	height m	95% CI	σ mm
R1	2.455	[2.437, 2.473]	9.0
R2	3.060	[3.039, 3.081]	11.0
R3	3.028	[3.007, 3.049]	11.0
R4	2.361	[2.343, 2.379]	9.0
R5	2.397	[2.379, 2.415]	9.0

Repeat spread cannot be measured (no second capture looks at the ceilings); the gate row is therefore *unverified*, not passed. Within-capture plane-fit σ is shown.

Damage (digitally staged, single_room_staged)

recall 1.00 (2/2), false positives 1; clean capture false positives 2.

Timing (2-core CPU, seconds)

run	reconstruct	layout	damage	total
lidar/single_room	32.8	1.6	149.8	184.5
lidar/floor_only	120.8	7.9	221.7	350.8
lidar/with_ceiling	403.3	13.8	345.9	763.3
video/single_room	44.4	1.3	—	46.0
photo/single_room	28.1	0.5	—	29.0
video/floor_only	195.9	8.3	—	204.5
photo/floor_only	32.5	0.7	—	33.6
video/with_ceiling	384.0	19.0	—	403.4
photo/with_ceiling	36.9	0.6	—	37.9

Video odometry, measured against ARKit

The video tier’s plans disagreed with LIDAR, so its RGB-only camera path was scored against the ARKit poses it never sees (bench/video_odometry.py). Three causes, each confirmed by measurement (DECISIONS D22): the depth model reads 1.26–1.35× too far (fixed by a per-device scale, leave-one-out in the benchmark); 17% of 0.3 s steps lose every track in fast turns (re-decoded at the native frame rate); rotation drifts between those gaps. Absolute rotation from image lines was built and ablated — doors and close-up fixtures mislead it, so it ships off. Means over the captures every configuration ran on (single_room, floor_only): (full per-capture table: bench/results/video_ablation.md)

configuration	mean ATE m	mean abs footprint err %
baseline (raw depth scale)	4.268	41.7
+ depth scale (leave-one-out)	3.384	72.8
+ fast-turn bridging	1.282	32.0
+ Manhattan rotation (lines)	1.675	30.6
+ fuse trusted keyframes only	1.675	31.9
+ floor-normal veto 15 deg	2.068	80.8

6. Calibration

Uncalibrated (sensor/fit/scale/drift budget only), 59% of repeat-capture wall differences fell inside the combined 95% intervals — overconfident. bench/calibrate.py fits a per-face residual on repeat captures (gross mismatches > 30 cm — a different wall, not noise — excluded: 15 of 37) and validates leave-one-pair-out: held-out floor_only~with_ceiling: 100% (n=12); held-out floor_only~single_room: 80% (n=5); held-out with_ceiling~single_room: 100% (n=5). Fitted per-face residual 6.5 cm (ambiguity weight 0.0). The intervals in every JSON are therefore the repeatability we can demonstrate, not the sensor spec. **Mono-depth scale (video/photo):** model depth / LiDAR depth = single_room 1.351, floor_only 1.2623, with_ceiling 1.3057; shipped correction 0.7655; each capture is benchmarked with the scale fitted on the other two (residual single_room +5.2%, floor_only -5.0%, with_ceiling -0.1%). Thin-tier intervals are set from those measurements (D24); the LiDAR footprint falls inside the thin tier’s 95% interval in 3 of 6 runs — where it does not, the error is registration/odometry, which the interval does not claim to cover (§5).

7. Fix loop

See docs/FIX_L00P.md (declaration committed before the fix; before/after regenerable with bash bench/fix_loop/run.sh before|after). Worst gate: LiDAR repeatability, 0% of walls within 1 cm / 0.5%. Iteration 1 (capture-invariant room splitting at doorways): 0 → 16.7% / 23.1% on two pairs; median-error prediction badly wrong (post-mortem: a second, systematic cause). Iteration 2 (furniture front vs wall: camera-oriented normals, outermost floor-reaching face): median |ΔL| 24.9 → 9.2 cm, within-3% 27.8 → 31.6%, but the declared 1 cm metric fell to 5.3% — the remaining error is within-capture drift duplicates (6–10 cm double walls), now attacked by the re-tuned drift correction (D16). Gate still fails; the report says why.

8. Head-to-head

Not producible from the sample data: the comparison needs a consumer app (e.g. magicplan, Polycam) run *in the same rooms*. The pipeline side is ready (bench/compare.py compares any two plans dimension by dimension; an app export converted to our JSON drops straight in).

9. Known failure modes (and what the output does about them)

- **Mirrors** (bathroom): LiDAR returns the mirrored room behind the glass → phantom free space; damage orthomosaics on that wall are reflections. Openings found on a wall are not filtered for mirror symmetry yet.
- **Glass shower screens / glass doors:** partly transparent to LiDAR; walls behind them are seen through reflections → damage detection unreliable there (a first staging in the bathroom, behind the screen and opposite the mirror, scored 0/2; kept as a failure case, DECISIONS D19).
- **Wet-look / glossy floors:** specular dropouts in depth; floors are not assessed for damage.
- **Low light:** LiDAR unaffected; mono-depth and KLT degrade — the video tier falls back to ICP/constant velocity (counted in capture.odometry) and its intervals are scale-dominated anyway.
- **Furniture against walls:** a face that stops short of the floor is not accepted as the wall when a floor-reaching face exists behind it; a full-height wardrobe is indistinguishable from a wall.
- **Open-plan junctions** (> 1.25 m): consistently merged into one room — a homeowner may draw two.
- **Ceiling never looked at:** reported as a prior, not a measurement.
- **Fast camera turns (video):** tracking at 10 fps breaks; the gap is re-decoded at the native frame rate, but turns of 40°+ in 0.3 s still fall back to constant velocity (counted in capture.odometry).
- **Few photos per room (photo):** top-view correlation needs overlapping structure; rooms with little overlap are placed on top of each other — the whole-property footprint is then far too small (§5).